PL-Series PLASMA LASER FILTER UNITS

Technical Specifications:

VANTERM

DGUV Test
IFA 1605032
Gefahrstoffgeprüft
Schweißrauch-
abscheideklasse W3



PL-4000 Plasma Laser Filter Unit W3 Certified.

W3 | EN ISO 15012-1 η≥ 99% IFA Certified | W3

H14 EN 1822 Hepa 14 certified 0,12 µm %99,997

IFA-DGUV Germany
W3 certified.
Cert. No: IFA 1605032

M DIN EN 60335-2-69 Dust Class M 0,2-2 µm %99,9 M

CE

PLUG-n-PLAY
Ready to operate!

Intended Use

- PL-Series filter units are specifically designed for extraction and dedusting of dust, fume and smoke generated during plasma, laser and OXY cutting operations.
- Plasma and laser cutting applications requires special design and unique features which cannot be satisfied by regular dust collectors because
  - Cutting fumes are essentially very fine particles of ave. 0,15 microns which may cause immediate filter clogging at regular dust collectors.
  - Filter cleaning and regeneration are very difficult to achieve for sustainable operation and reasonable spare filter replacement and maintenance intervals.
  - There is a very high risk of fire, due to smoldering hot particles that are generated during thermal cutting operation if proper precautions have not been held.
- VANTERM PL-Series filter units with W3 dedusting Class certification and filter elements average lifetime of 20 000 hours are proven their fitness for the job with thousands of applications all over the world.
- Thanks to the HEPA 14 filter quality of VANTERM Panel Filter Technology, PL Dust collectors separate the ultra-fine dust particles including as small as 0,12 microns with 99,997% filtration efficiency.

Operation Principle

- Dust laden air is extracted from cutting bench of laser, plasma or Oxy cutting unit. First, dust laden enters the pre-separator chamber where hot and big particles are separated from main stream aerodynamically. Afterwards, dust laden air is filtered at the panel pleated filters. Clean air is extracted through fan-silencer group and returned clean as VANTERM PL-Series are W3 certified.
- When K-type VANTERM filter cleaning system senses that filters are needed to clean, Jet-pulse filter cleaning is activated in order to clean panel pleated filters elements. A sudden burst of pressured air is released to clean filter by freeing the pores of the filter media and shaking of dust cake on the elements.

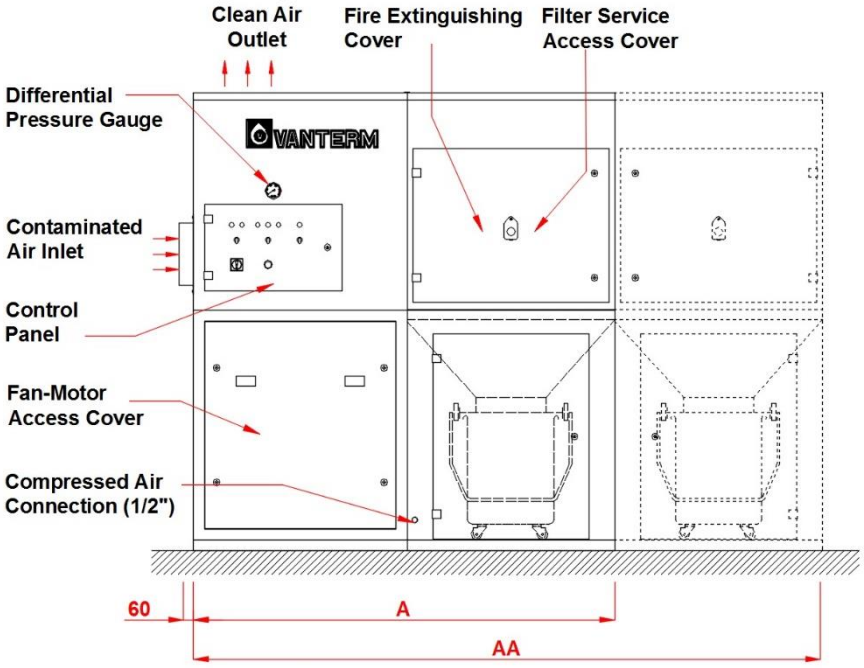
Standard Features

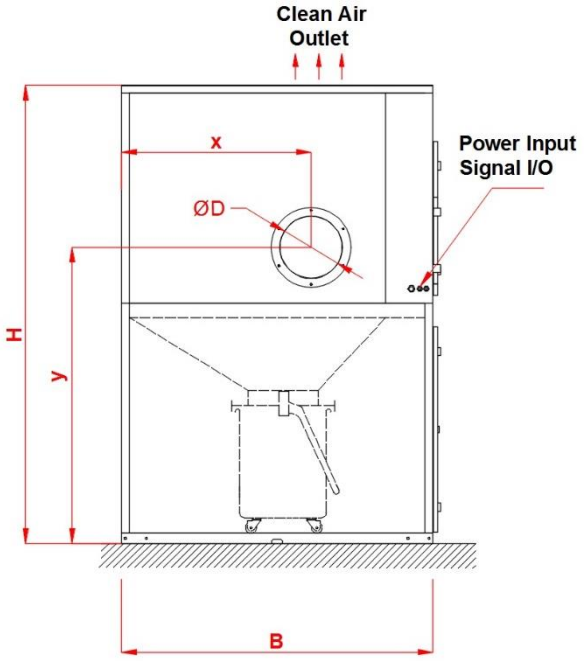
All PL-Series Filter Units are W3 Certified and delivered ready to operate as standard.

- **Dedusting Class:** W3 according to EN ISO 15012-1
- **Power Input:** 480V ~ 3P PE - 60 Hz.
- **Pre-Separator:** Aerodynamic Spark and Big particle separation in order to minimize fire risk.
- **Fan:** Aluminum High Efficiency Radial Backward Curved G2.5 balanced with Direct Coupling following ERP 2015.
- **Filter Type:** Panel Pleated Surface Filtration Dust Class M and H14 ePTFE membrane Polyester Non-woven.
- **Filter Efficiency:** HEPA14 certified acc. To EN 1822 standard – Filter Efficiency Over 99,997% of 0,12 micron particles. Filtration degree exceeds MERV 16 rating.
- **Filter Life time:** Average 20 000 h state-of-the-art filter lifetime above industry standards with cutting edge filter design, media and cleaning system.
- **Filter Cleaning:** VANTERM K-type fully automatic Differential Pressure (ΔP) controlled with high efficiency solo jet-pulse cleaning.
- **Pressured Air Input:** min.6 bar / 87 PSI (Reduced to 4,5 bar / 65 PSI internally with built-in regulator)
- **Sound level:** Extreme Low sound level below ≤69 dB(A) with built-in silencer after fan.
- **Operation Mode:** Panel Operation and slave mod (with NO relay connection from parent) as standard.
- **Output Signals:** Digital Motor Fault, Normal Operation and Low Air Flow Signals
- **Dust Bin:** Easy to Handle locking mechanism 80 to 160-liter capacity.

Optional Features

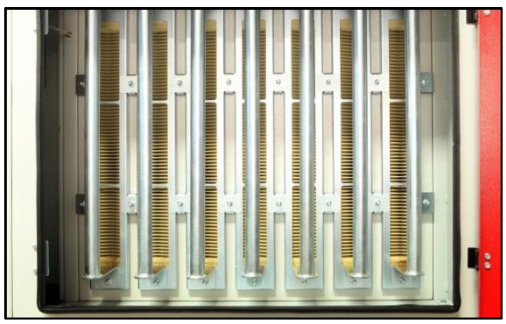
- Any three phase Voltage and frequency (ex.400V ~ 3P PE - 50 Hz) configuration is available upon request.





TYPE	CAPACITY (CFM)	MOTOR (HP)	FAN PRESSURE (Pa)	FILTER AREA (ft²)	A (in.)	AA (in.)	B (in.)	H (in.)	ØD (in.)	DUST BIN (liter)	X (in.)	y (in.)	WEIGHT (lbs.)
PL-1000	590	2,0	3 000	215	72.6	-	49.6	72.5	5.90	21	29.8	47.2	992
PL-1500	885	3,0	2 950	269	72.6	-	49.6	72.5	6.30	21	29.8	47.2	1036
PL-2000	1 180	4,0	3 000	377	77.1	-	49.6	72.5	7.87	21	29.8	48.5	1080
PL-2500	1 470	5,5	3 300	430	82.0	-	49.6	72.5	8.66	21	29.8	48.5	1158
PL-3000	1 770	5,5	2 900	614	74.3	-	59.6	86.2	9.84	21	35.5	55.9	1268
PL-4000	2 360	7,5	3 100	716	79.2	-	59.6	86.2	11.81	21	35.5	55.9	1327
PL-5000	2 950	7,5	2 750	915	89.0	-	59.6	86.2	11.81	21	35.5	55.9	1540
PL-6000	3 540	10,0	2 950	1023	93.9	-	59.6	86.2	13.78	21	35.5	55.9	1753
PL-8000	4 720	10,0	2 300	1432	-	118.3	59.6	86.2	15.75	42	35.5	55.9	2039
PL-10000	5 900	15,0	2 600	1636	-	133.1	59.6	86.2	17.72	42	35.5	55.9	2249
PL-12000	7 080	20,0	2 850	1841	-	143.0	59.6	86.2	17.72	42	35.5	55.9	2557

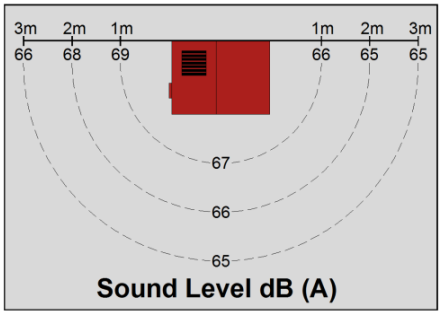
***Ongoing product development may subject the technical specifications to alteration without notice.**



Jet-pulse Cleaned Panel Filters are H14 certified.



Easy Dust Bin Mechanism



Low Sound Level with Silencer

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